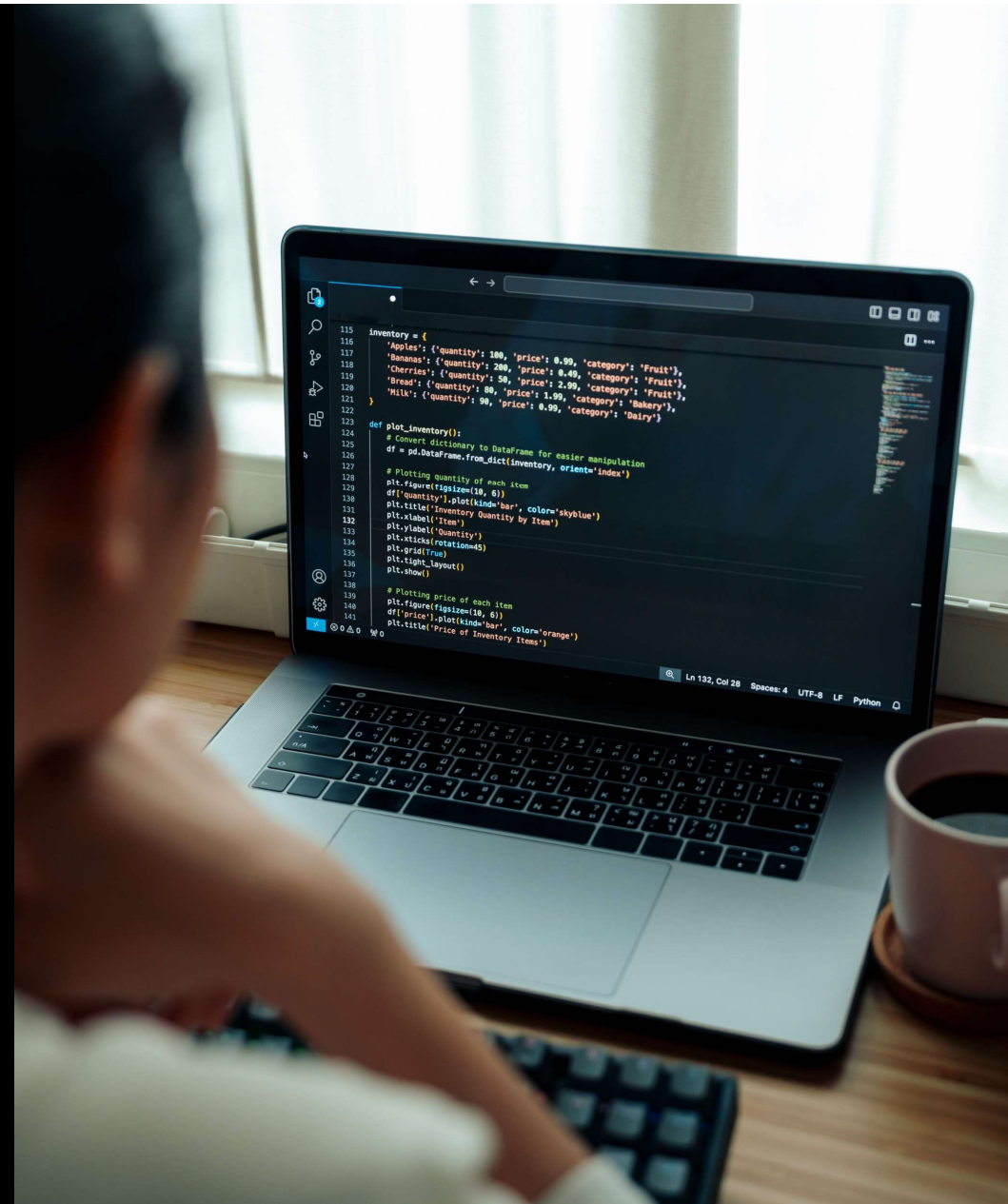


Module 1: Introduction and core IT principles



Overview

Day 1

- Diversity and cultural awareness
- Communication in the workplace
- Health and safety
- Numbering systems

Day 2

- Technical documentation and problem solving
- Introduction to STARRS
- Task prioritisation
- Introduction to Cisco Packet Tracer
- Troubleshooting activity

How to complete this course

This module covers practical's using Cisco Packet Tracer, GoToMyPC / LOD, and access to Miro.

Please ensure you have these installed and working on your own machine.

If you have any issues, please inform your tutor.



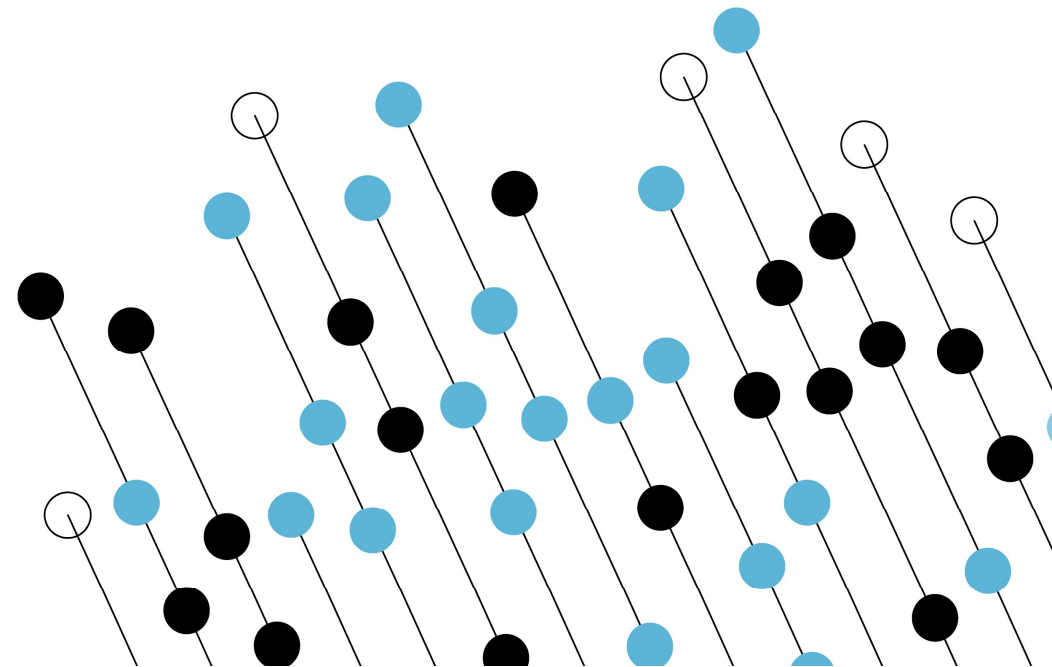
How to complete this course

As part of your Live event, you will have the slides and activity book to complete.

- Open your workshop worksheet for this module and use it to document your answers.
- Add your full name and date in the document header.
- Use this PowerPoint as a guide to take you through the course.
- Where possible, refer to the business you work for in your answers.
- If you need help, message or email your tutor.



Diversity and cultural awareness



Diversity and cultural awareness

- Diversity and culture could be summarised as the things we share, where we come from and, our family values / beliefs.
- They are drawn from several influences:
 - Age
 - Gender
 - Race
 - Religion
 - Language / dialect
 - History
 - Dress
 - Festivals and celebrations

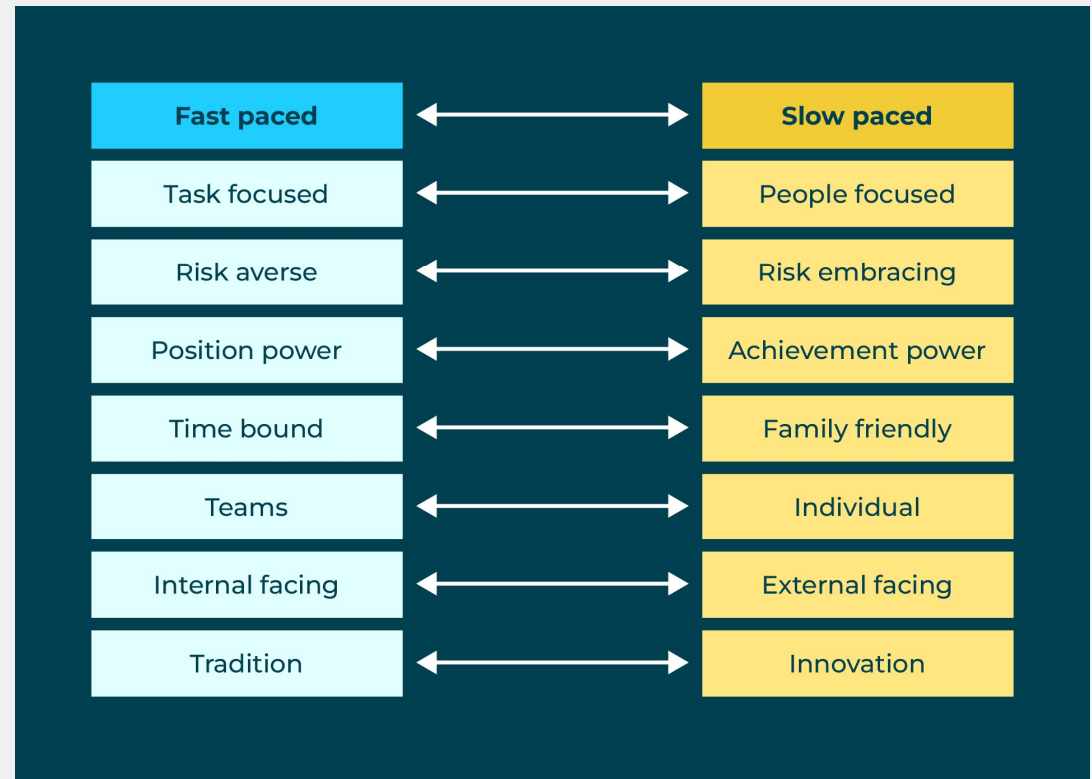
Organisational culture

Think of a culture code as the employee handbook. It should include:

- your mission statement.
 - core values.
 - how to work and live by your values.
 - any company traditions.
 - any additional shared beliefs.
-
- If your culture and values don't align, your employees, customers, and bottom-line performance may suffer.
 - Organisations create values statements that will help align employees and organisational culture to drive performance.

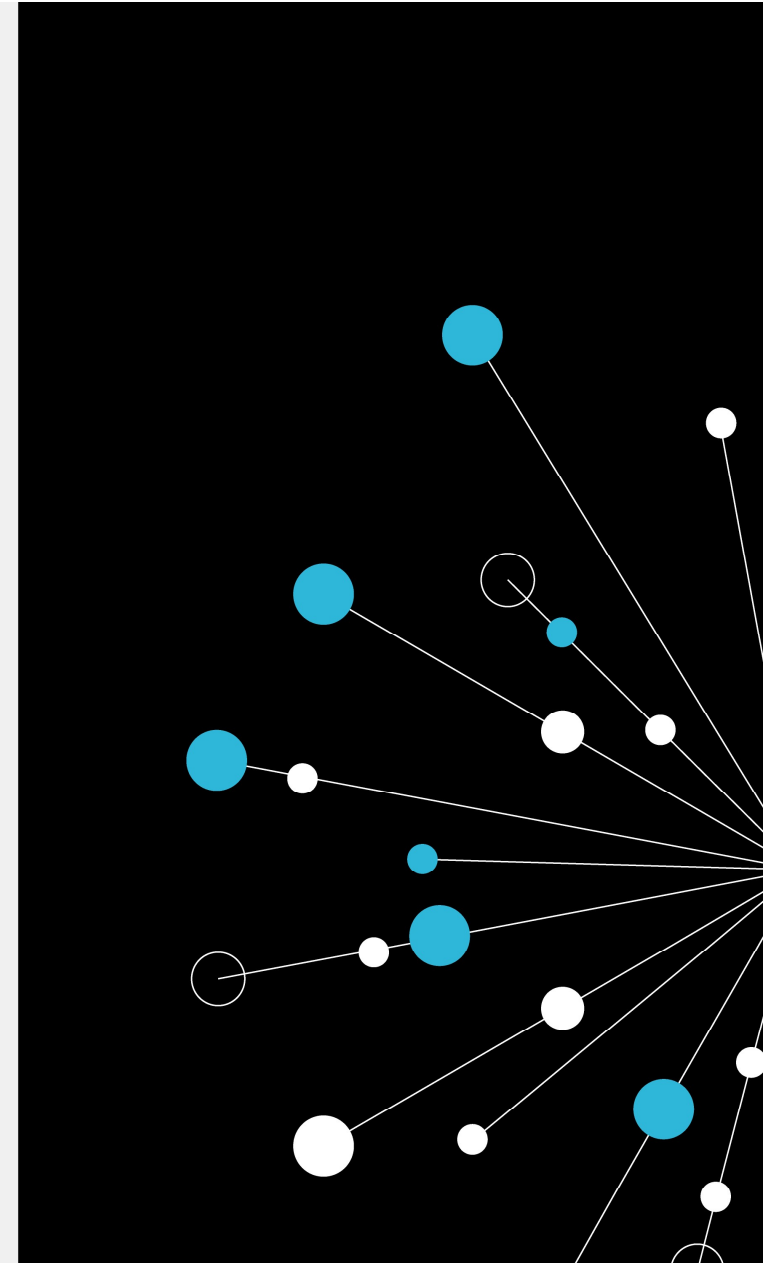
Culture cues

- Not all IT sectors exhibit the same cultural dynamics.
- Some industries, like finance, may be fast-paced, while others, such as education, might operate at a slower pace.
- Consider the industry you work in and assess how it aligns with these different cultural tempos.

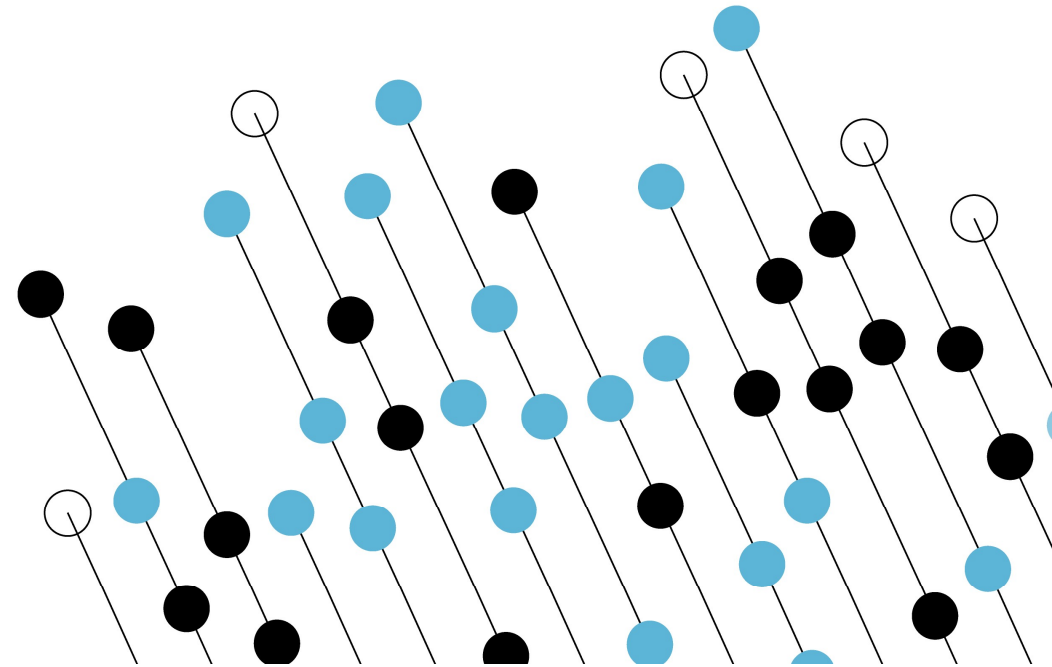


Diversity and cultural awareness

1. Define vision, mission, and value statements. Find examples online from your favourite brands or your workplace.
2. List the benefits of working in a diverse organisation.
3. Describe your workplace cultural cues.



Communication in the workplace



Communication in the workplace

An Information Communications Technician is a highly technical role. The number of acronyms and technical jargon you use without even thinking can be overwhelming for a non-technical person.

**'Can you click
on Start for
me?'**

**'Press the
Windows key
on your
keyboard.'**

**'Type in
IPCONFIG.'**

**'User requires
Bit Locker
Recovery Key.'**

Communication in the workplace

- Adjusting how you communicate is a skill and takes practice to help make your customer feel at ease.
- Your written and verbal skills can be significantly challenged when trying to simplify a complex subject.
- Considering how to disseminate the information so that it is well received can be a challenge.
- Oversimplify and you become patronising.
- Under-simplify and you could appear like you are talking another language.

Communication in the workplace

Two principles to refer to are:

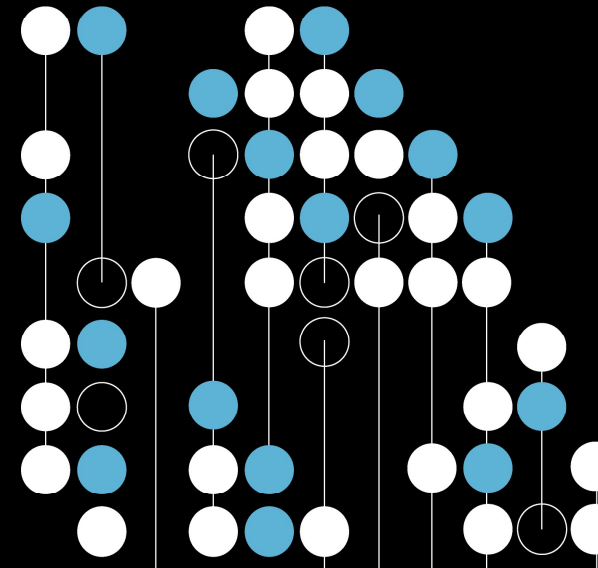
1. **KISS:** Keep It Short and Simple
2. **DRY:** Don't Repeat Yourself

Best practices to follow are:

- consider your audience.
- provide enough context around the task.
- avoid using complicated or unusual words.
- avoid unnecessary repetition and assumptions.
- show appreciation and empathy.
- start your sentence with the main topic.

**What methods of
communication do you use in
your workplace?**

QA



Activity 2: Communication in the workplace

1. Discuss the advantages and disadvantages of using the following communication methods for service desk tickets in the workplace:

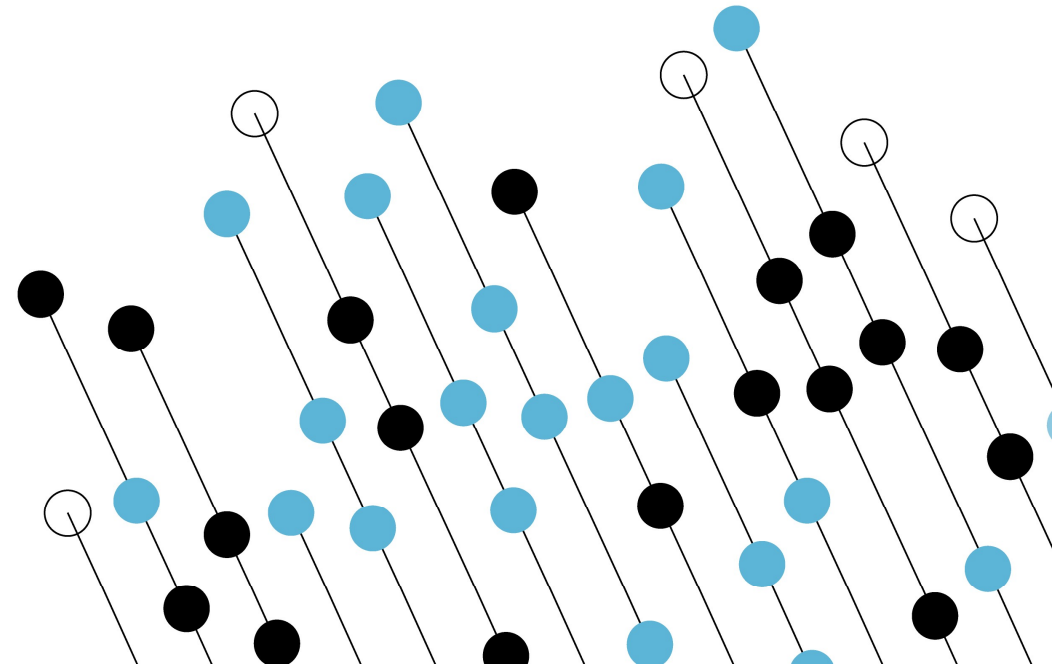
	Advantages	Disadvantages
Phone call		
Instant Messaging		
Email		

2. Respond using clear steps to resolve this customer issue:
 - 'Hello IT Team, my computer is not working. I'm unable to go to any websites, Teams is also not working, I can't access the intranet or shared drive. I've restarted many times but still nothing is being connected. Could you give me a new machine please?'

Communication in the workplace

- Communications is a technique you will practice throughout your career and is one that you will continually adapt.
- The art of good, clear dialogue can save you a considerable amount of time and frustration.
- It is well worth reflecting on your written or verbal conversation to note what did / didn't work well.
- This will enable changes to be made for the future.

Health and safety



Health and safety

- The Health & Safety at Work Act 1974 (HASAWA) lays down wide ranging duties on employers.
- Employers must protect the 'health, safety, and welfare' at work of all their employees and others on their premises, including:
 - temporary staff.
 - casual workers.
 - self-employed workers.
 - clients.
 - visitors.
 - the public.

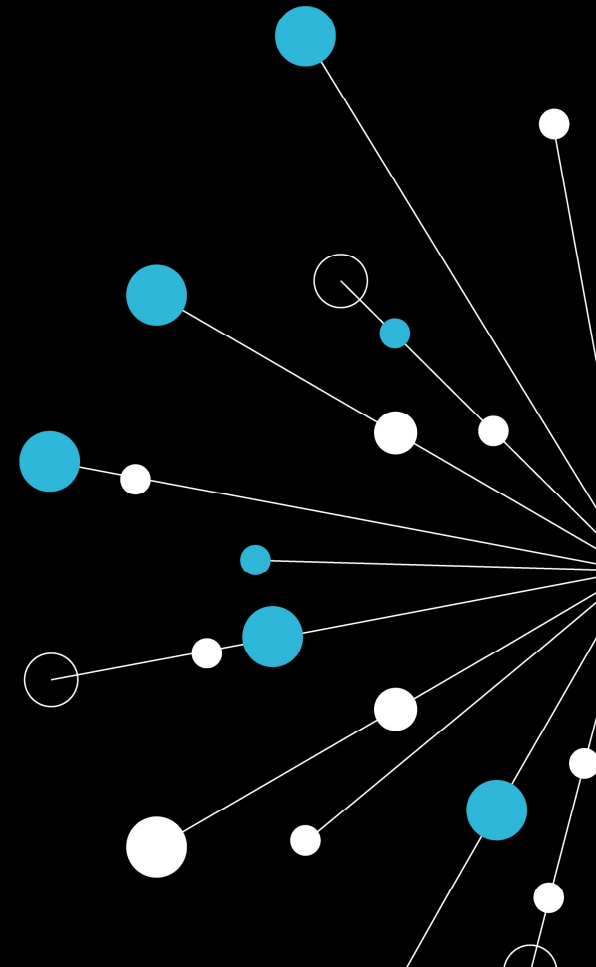
Health and safety

Employers have a duty, so far as is reasonably practical, to:

- ensure a safe place to work.
- provide safe equipment, plant, and machinery.
- provide safe and competent people working alongside you.
- carry out risk assessments as set out in regulations and take steps to eliminate or control those risks.
- inform workers about potential hazards associated with any workprocess, chemical substance, or activity, including providing instruction, training, and supervision.
- appoint a competent person responsible for health and safety.
- consult with workplace safety representatives.
- provide adequate facilities for staff welfare at work.

Activity 3: Health and safety

1. List and describe the responsibilities you have to ensure health and safety is followed in your role.



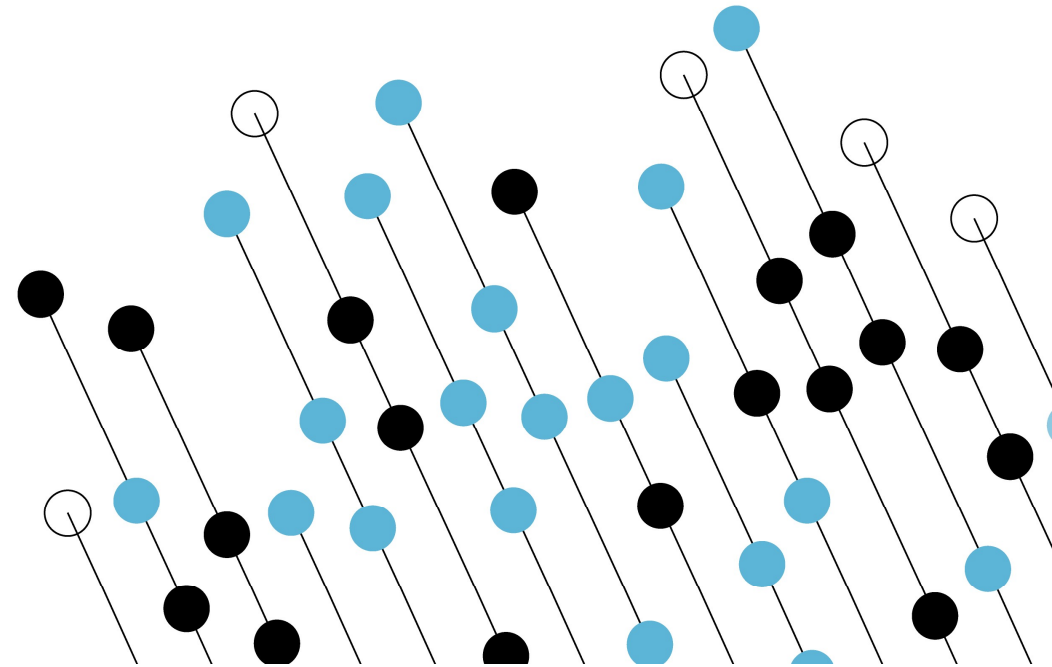
Health and safety

How is health and safety evolving around technology?

- Offsite construction
- Drone tech
- Smart buildings
- Smart sensors and connected assets
- Wearable tech
- Virtual training
- Fatigue-detection technology
- AI advancements



Electricity at Work Regulations



Electricity at Work Regulations

The Electricity at Work Regulations 1989 expand on the rules regarding electrical safety in the Health and Safety at Work Act 1974 and cover several areas that electricians need to know.



Electricity at Work Regulations

Portable Appliance Testing, called PAT testing, is used to test all electrical devices.

These tests must be completed regularly, usually annually, and must be completed by trained personnel.

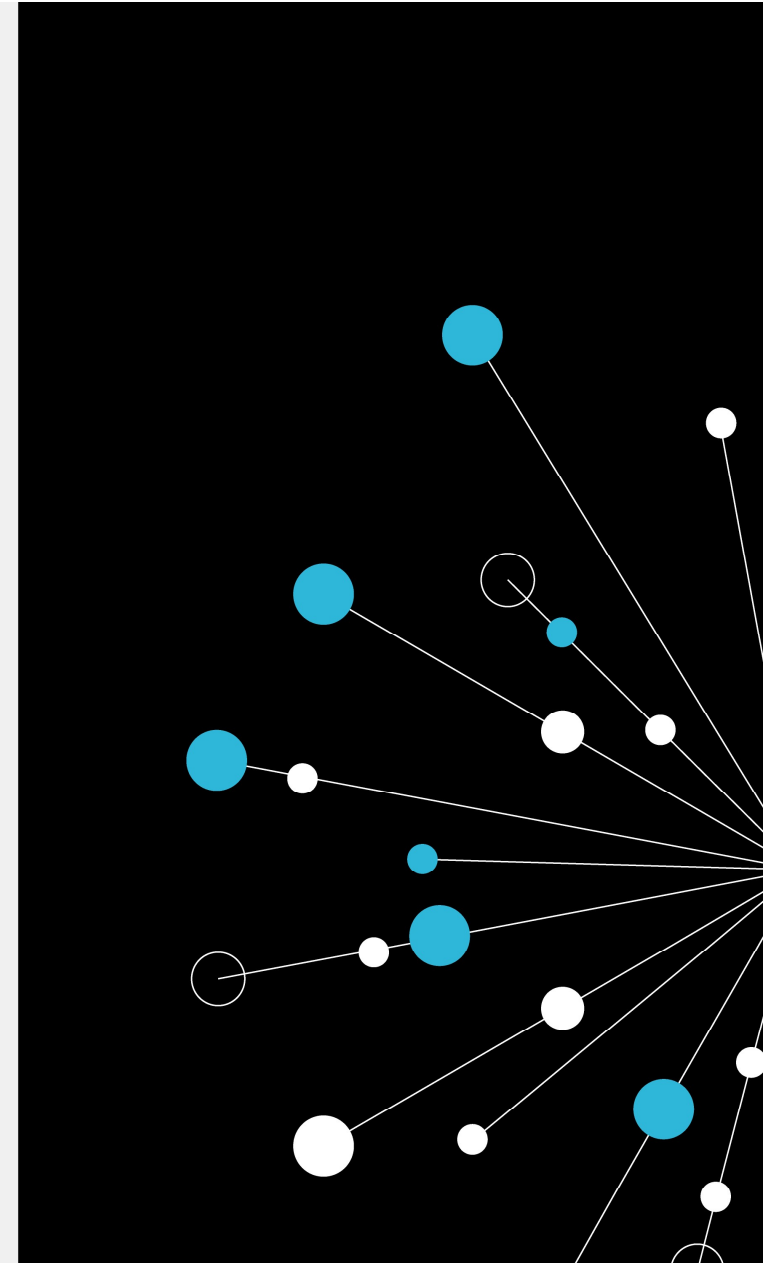
The three parts of the test include:

- visual inspection.
- earth continuity test.
- insulation test.

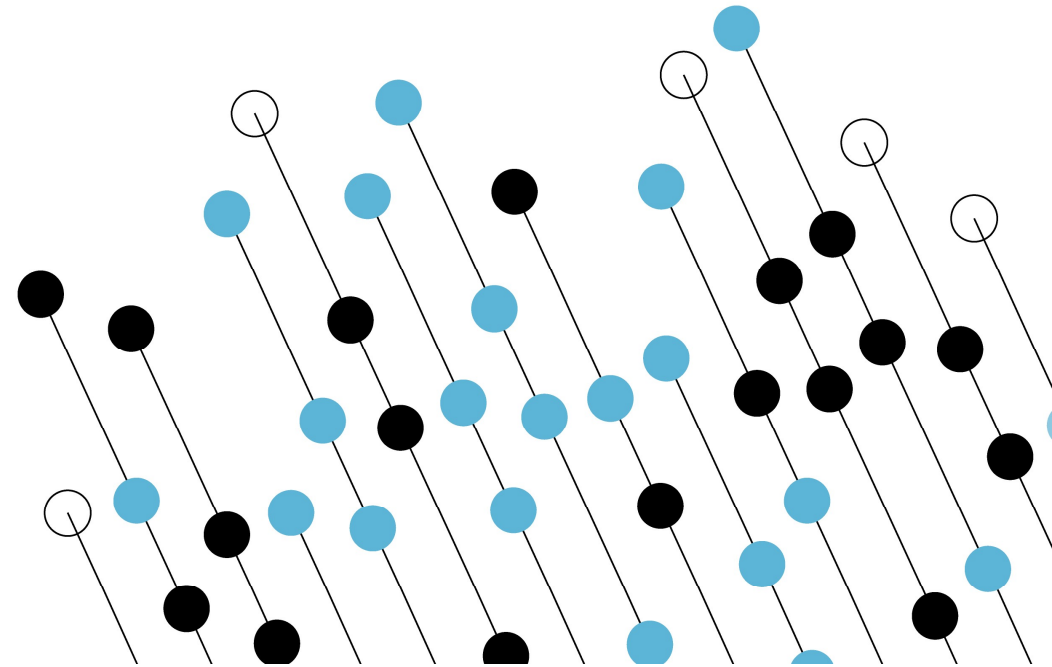


Activity 3: Health and safety

1. List and describe what electrical items would you consider to be prone to causing fires in your homes and workplace.



Manual Handling Operations Regulations



Manual Handling Operations Regulations

Your role may well involve lifting and carrying equipment in your workplace.

The Manual Handling Operations Regulations 1992 (amended in 2002) apply to a wide range of manual handling activities, including

- lifting.
- lowering.
- pushing.
- pulling.
- carrying.



Manual Handling Operations Regulations

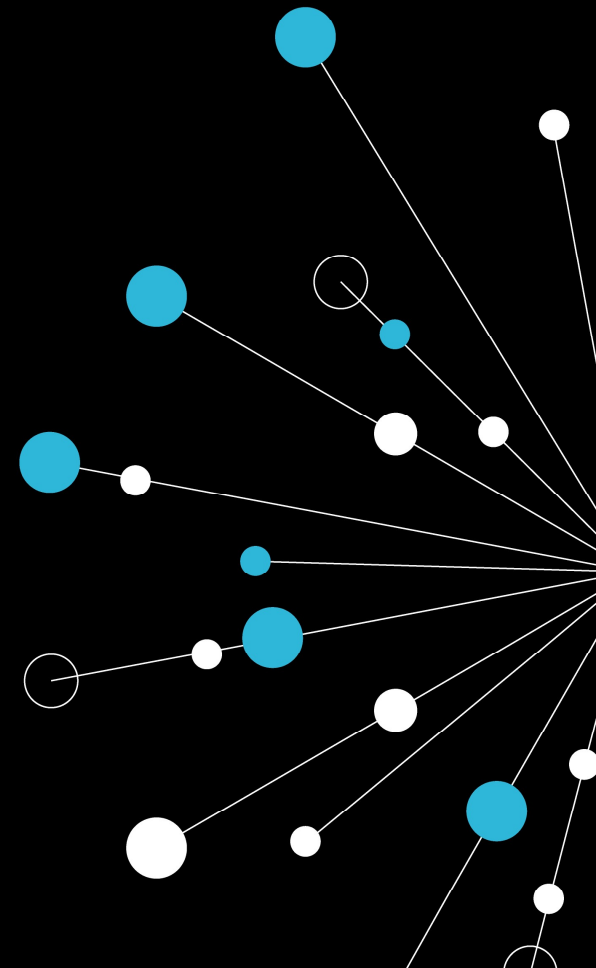
The MHOR 1992 set out a clear ranking of measures for dealing with risks from manual handling, these are:

1. avoid hazardous manual handling operations so far as is reasonably practicable.
2. assess any hazardous manual handling operations that cannot be avoided.
3. reduce the risk of injury so far as is reasonably practicable.

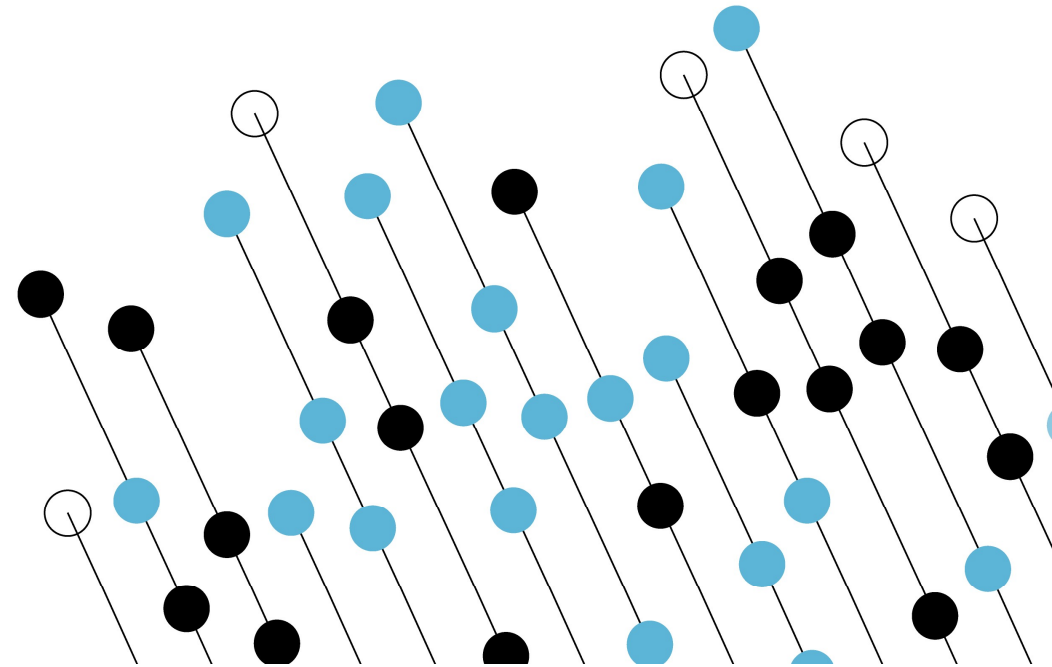


Activity 3: Health and safety

1. What prerequisites would you consider when handling / moving IT equipment in your workplace?



Display Screen Equipment Regulations



Display Screen Equipment Regulations

The Display Screen Equipment (DSE) Regulations 1992 only apply to employers whose workers extensively use computers and / or laptops as a normal part of their daily activities for continuous periods of more than one hour.

An employer with DSE users is responsible for:

- analysing the risk and ensuring controls are in place to reduce.
- providing DSE training.
- providing eyesight tests and special spectacles if required.
- reviewing the assessment when the user or DSE changes.



Display Screen Equipment Regulations

Training and information: Providing comprehensive training on safe DSE use is essential. This training should cover aspects like maintaining good posture, proper adjustment of chairs and desks, appropriate screen brightness and position, and the importance of taking regular breaks.

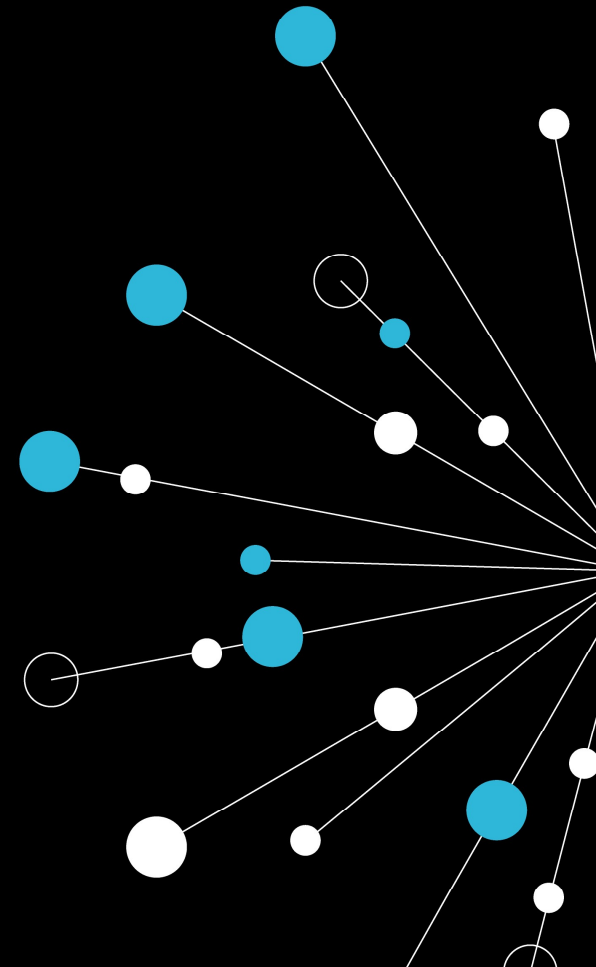
Preventing Repetitive Strain Injury (RSI): Employers should focus on RSI prevention by designing ergonomic workstations and promoting good working practices, both in the office and for remote workers.

Ergonomic equipment: Choosing the right equipment is vital. For instance, if laptops are used frequently, providing docking stations with separate keyboards and mice can help maintain ergonomic standards.

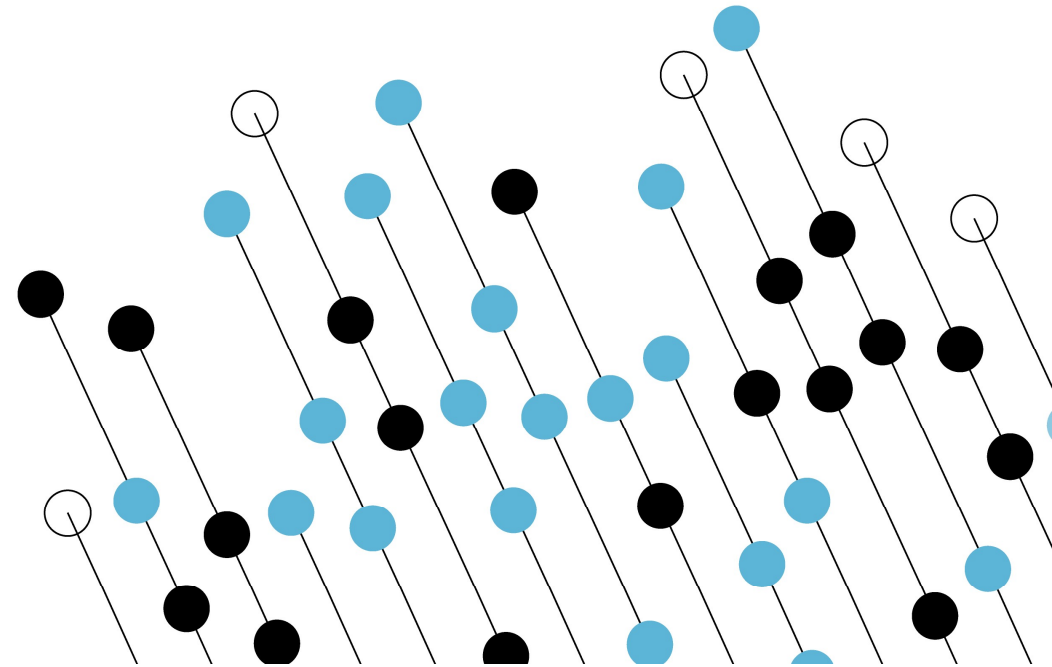


Activity 3: Health and safety

1. List and describe DSE equipment that would help assist your role within your workplace.



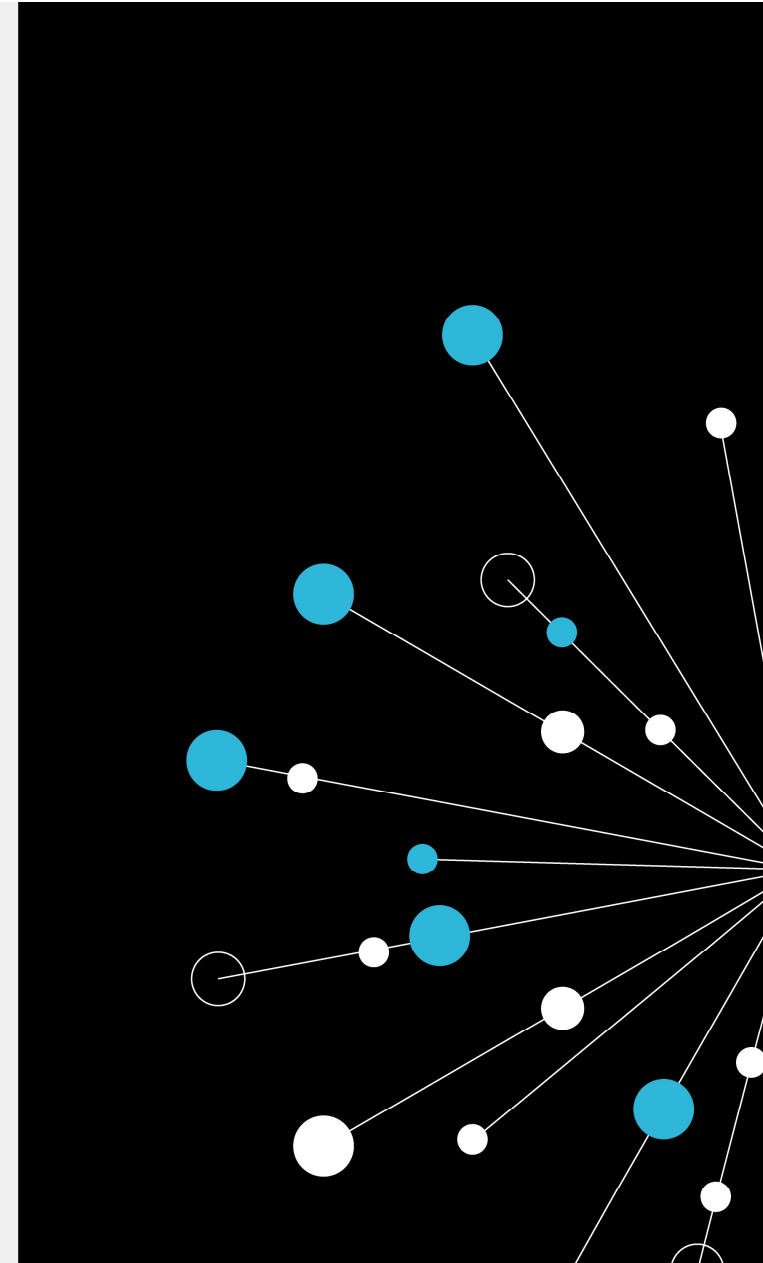
Numbering systems



Knowledge Check

What numbering system does binary use?

- A. Base-10
- B. Base-2
- C. Base-16



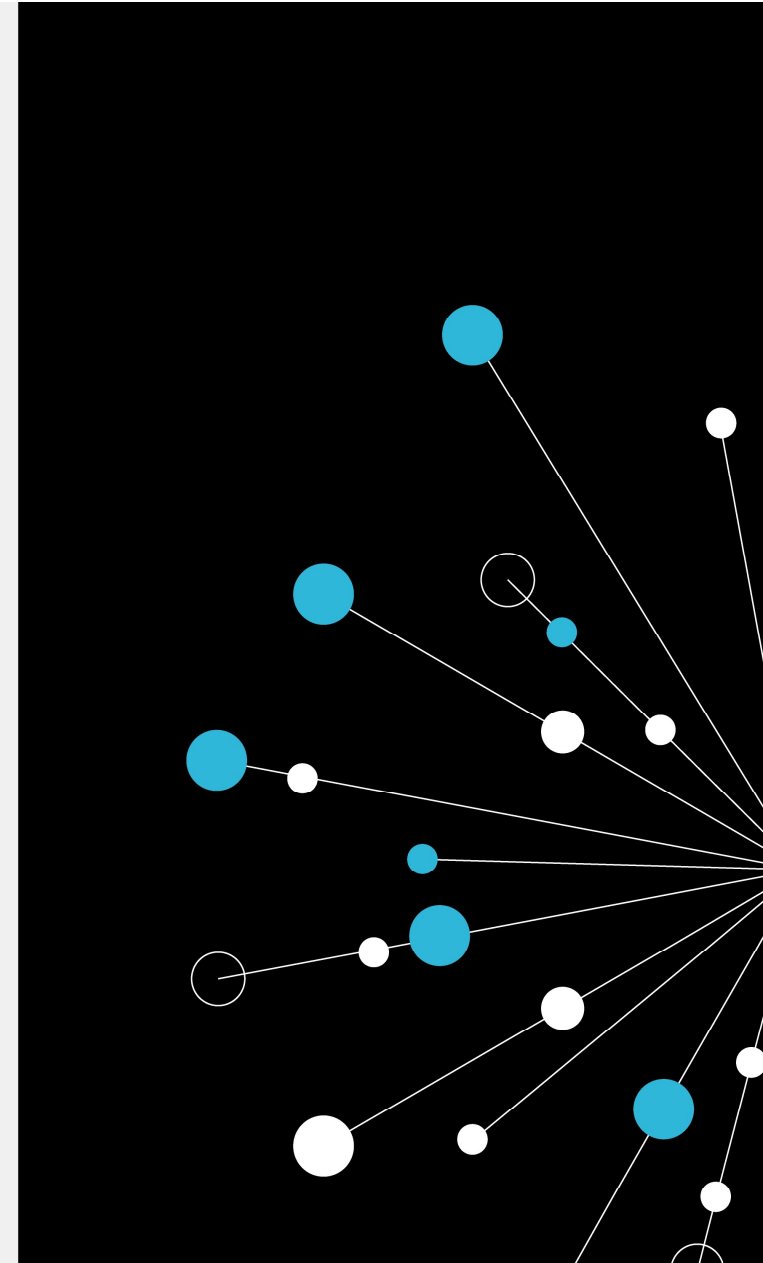
Knowledge Check answer

Question stem?

This correct answer is: Base-2

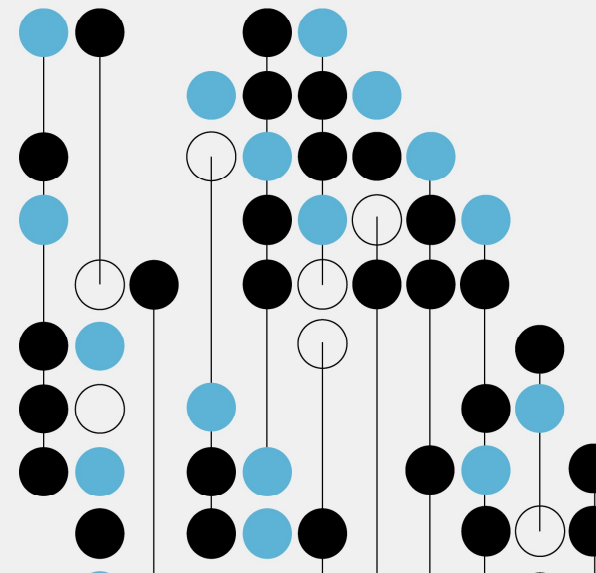
Binary uses the digits 0 and 1 represent numbers.

Decimal is a base-10 system, and hexadecimal is a base-16 system.



Decimal

QA



Decimal numbers

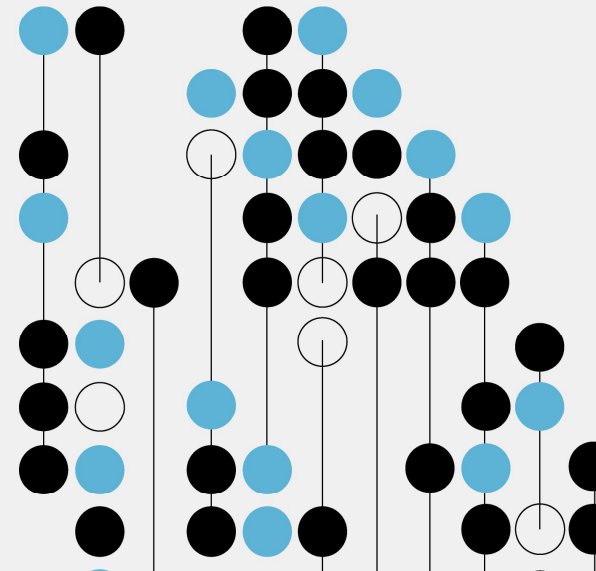
- Decimal numbers are numbers expressed in the base-10 numeral system, and includes ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.
- Each place value is a power of 10 (i.e., 10x), for example, in the number 345.67:

10^2	10^1	10^0	.	10^{-1}	10^{-2}
x 100	x 10	x 1	.	x 0.1	x 0.01
3	4	5	.	6	7

- Decimal numbers are widely used in everyday life for various purposes such as currency, measurements, and general calculations due to their simplicity and ease of understanding.

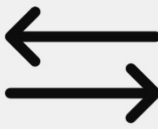
Binary

QA



Binary

- Binary is a base-2 numeral system that uses only two digits: 0 and 1.
- Each place value in a binary number is a power of 2. For example, the decimal number 13 in binary is the number 1101:

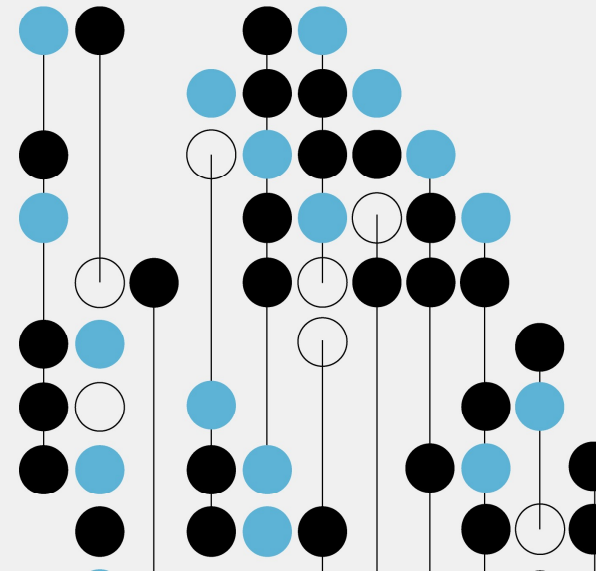
10^1	10^0		2^3	2^2	2^1	2^0
x 10	x 1		x 8	x 4	x 2	x 1
1	3		1	1	0	1

Binary

- Binary is used in computers and digital electronics because it aligns well with the two-state nature of electronic circuitry (on / off, true / false, high / low).
- Some key applications include:
 - **Computing:** All data and instructions in computers are represented in binary form.
 - **Networking:** IPv4 addresses, and subnet masks are often represented in binary.
 - **Digital logic:** Logic gates and circuits use binary for operations.

Hexadecimal

QA



Hexadecimal

- Hexadecimal (or hex for short) is a base-16 numeral system that uses sixteen distinct symbols to represent values.
- These symbols are the digits 0 to 9 and the letters A to F, where A stands for 10, B for 11, C for 12, D for 13, E for 14, and F for 15.

Decimal	Binary	Hex
0	0	0
1	1	1
2	10	2
3	11	3
4	100	4
5	101	5
6	110	6
7	111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Comparing numbering systems

Decimal

10^1	10^0
x 10	x 1
1	3

Binary

2^3	2^2	2^1	2^0
x 8	x 4	x 2	x 1
1	1	0	1

Hex

16^0
x 1
D

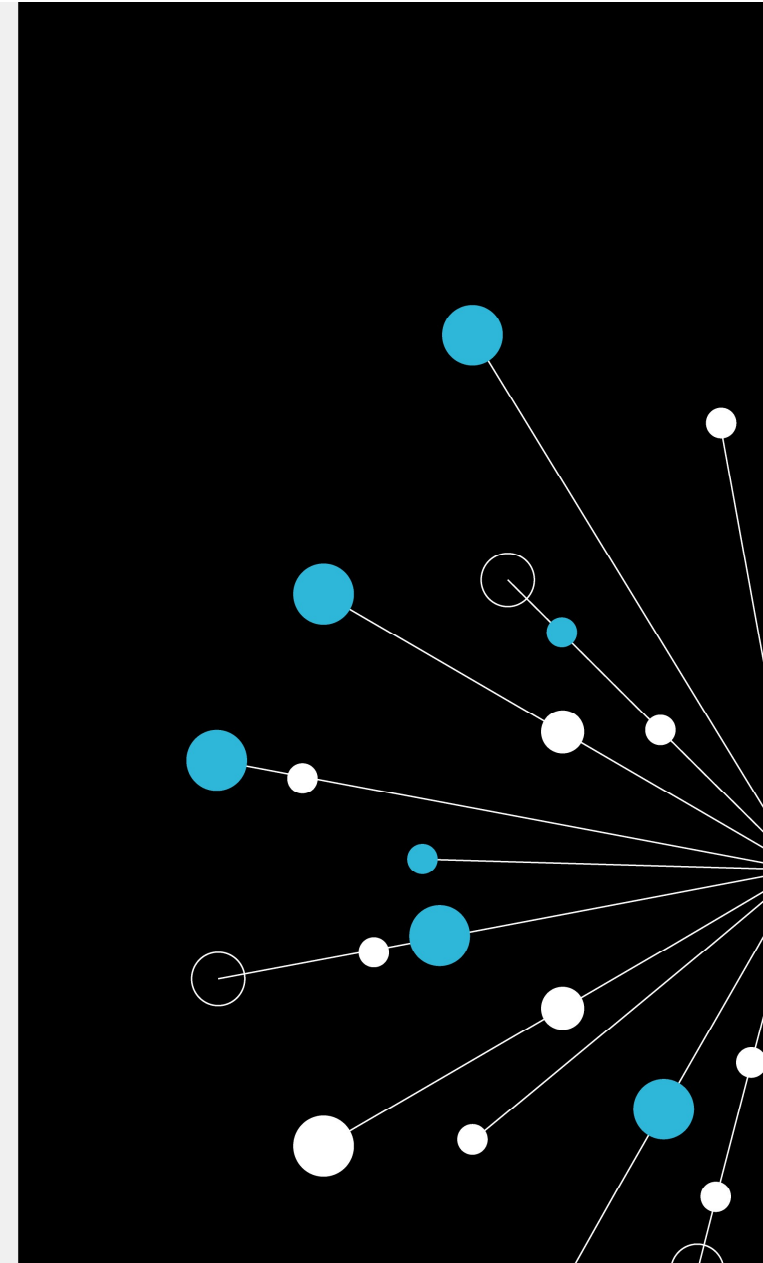
Hexadecimal

Hexadecimal is widely used in computing and digital electronics for various reasons:

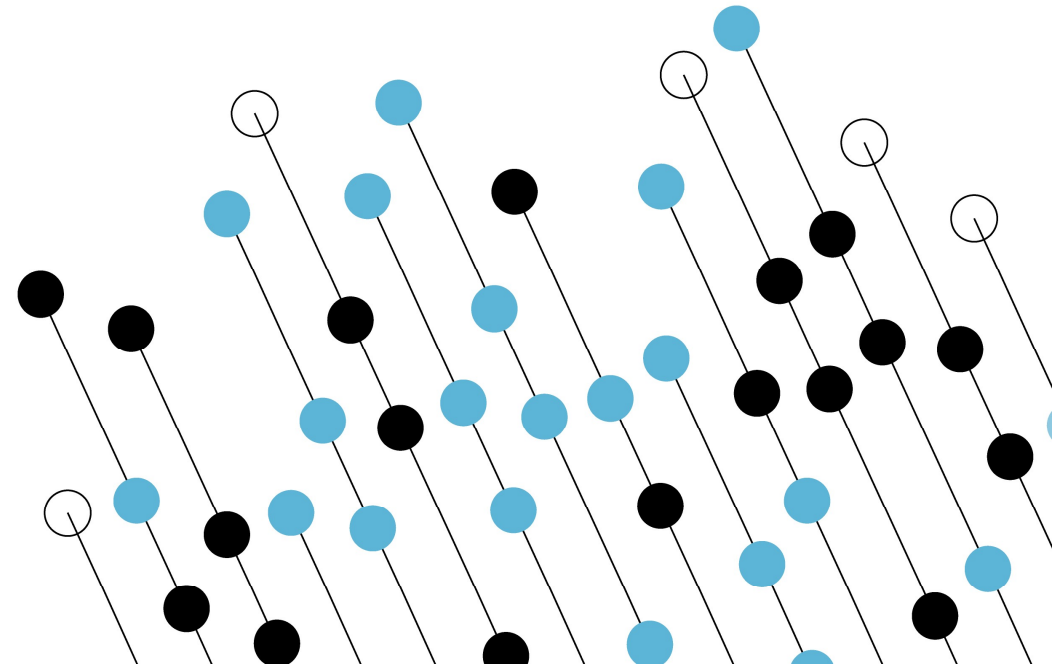
- **Memory addressing:** Hexadecimal notation is used to represent memory addresses (MAC and IPv6) in a more readable form than binary.
- **Colour codes:** In web design, colours are often represented in hexadecimal format, such as #FF5733.
- **Compact representation:** Hexadecimal provides a more compact and human-friendly way to represent binary numbers.

Activity 4: Numbering systems

1. Complete the conversion tasks.
2. What is the lowest and highest decimal number you can make in an octet?
3. How many bits are used to make an IPv4 address?
4. How many bits are used to make an IPv6 address?



Technical documentation and problem solving



Technical documentation

- Technical communication includes any form of communication that has one or more of the following characteristics:
 - communicating about technical or specialised topics, such as computer applications, medical procedures, or environmental regulations.
 - communicating by using technology, such as web pages, help files, or social media sites.
 - providing instructions about how to do something, regardless of how technical the task is or even if technology is used to create or distribute that communication.

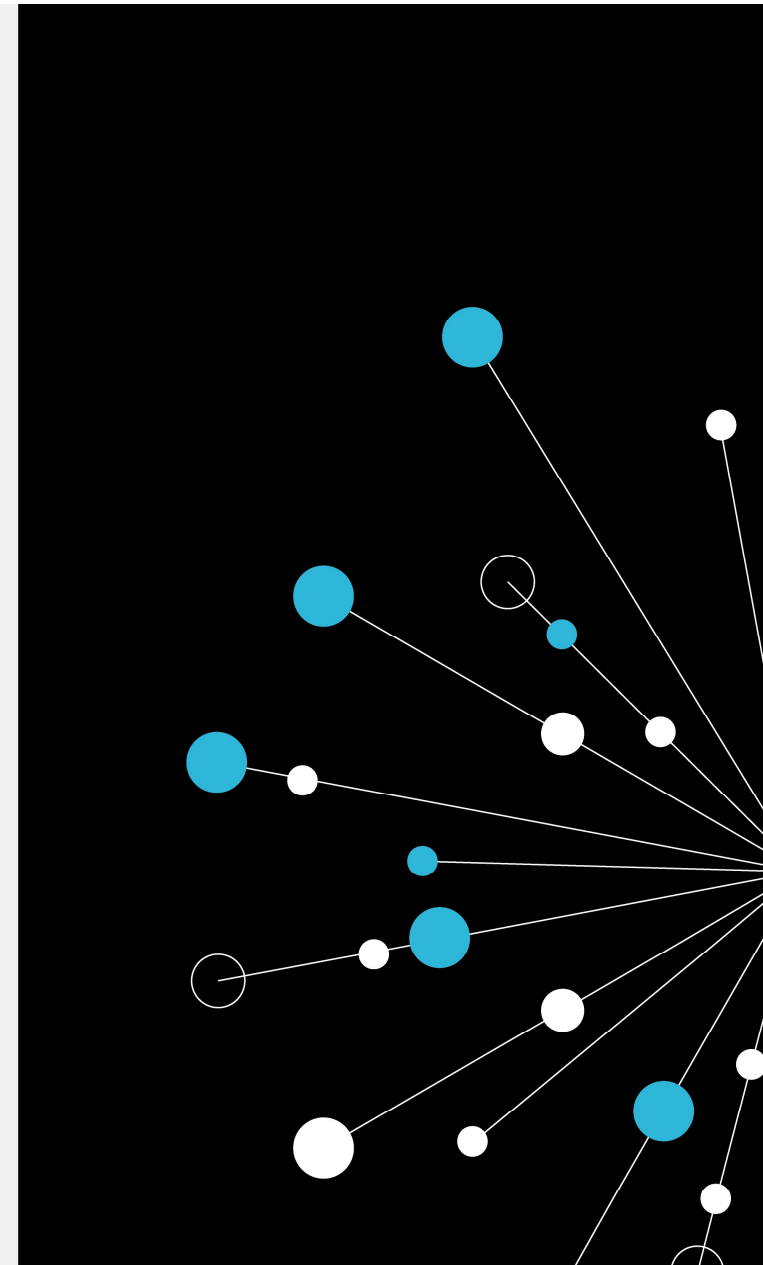
Technical documentation

- The purpose of technical writing is to convey complex information:
 1. clearly.
 2. accurately.
 3. concisely.
- This makes it easier for readers for to understand, use, or act upon the information given.

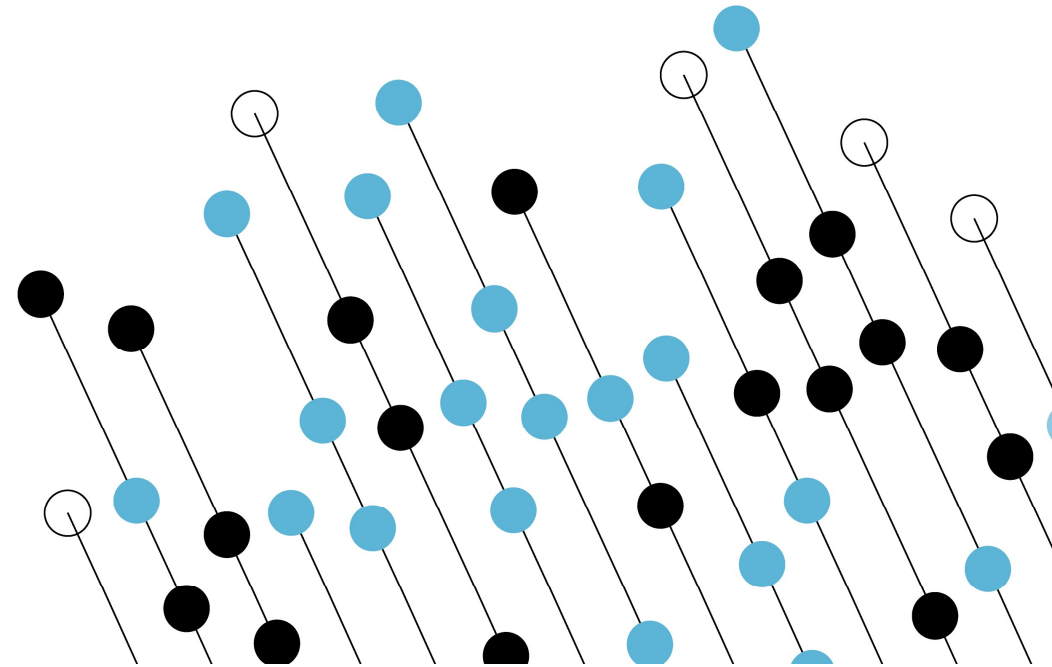
Activity 5: Technical documentation

1. You are working on the service desk and have been assigned a ticket.

Read through the ticket (in the workshop worksheet) and create a technical document for the customer.



Introduction to STARRS



S.T.A.R.R.S

The STARRS method can be adapted to many situations. It is easy to follow and can help to structure your answers in a job interview and your End-Point Assessment.

Situation

Task

Action

Result

Reflection

Strengthen

Situation

When describing the situation:

- explain the situation you were in.
- aim to answer the questions 'what / where / when'.

Task

When describing the task:

- what did you do, and what did the task or role involve?
- explain your tasks and responsibilities briefly.

Action

When describing your actions:

- what did **you** do to meet the task objective?
- how did you do it and what skills did you need?

Result

When describing the result:

- what was the outcome?
- did you make a difference?
- what evidence do you have?

Reflection

When reflecting:

- what did you do well?
- what didn't go so well?

Strengthen

When describing what you have learnt from the experience:

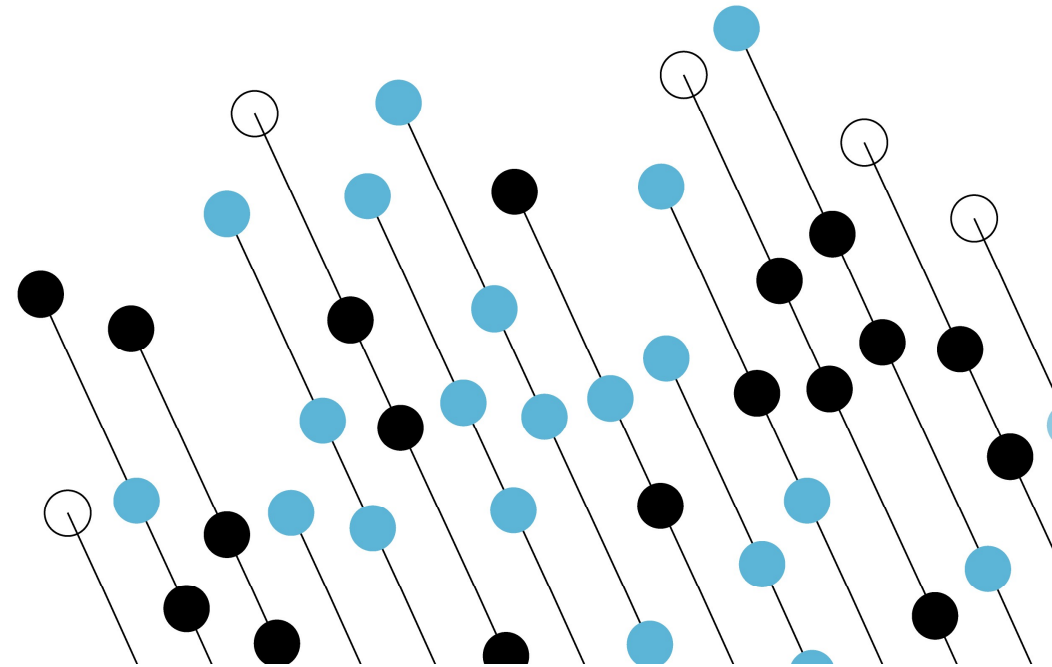
- what would you have done differently?
- what could you have improved?

Activity 6: STARRS

- The STARRS method can be adapted to many situations. It is easy to follow and can help to structure your answers in a job interview and your End-Point Assessment.
- Complete the table in the worksheet with the details:

Acronym	Key word	Explanation
S		
T		
A		
R		
R		
S		

Task prioritisation



Task prioritisation

- Creating a to-do list is the starting point for tackling your workload.
- But when you're short on time and can't accomplish everything in a single day, how do you decide what to address first?
- There are various task management tools out there that can help you in your personal life and work life to get your tasks and jobs done.

Task prioritisation

The Eisenhower Matrix is a simple tool for prioritising tasks by dividing them into four categories based on urgency and importance:

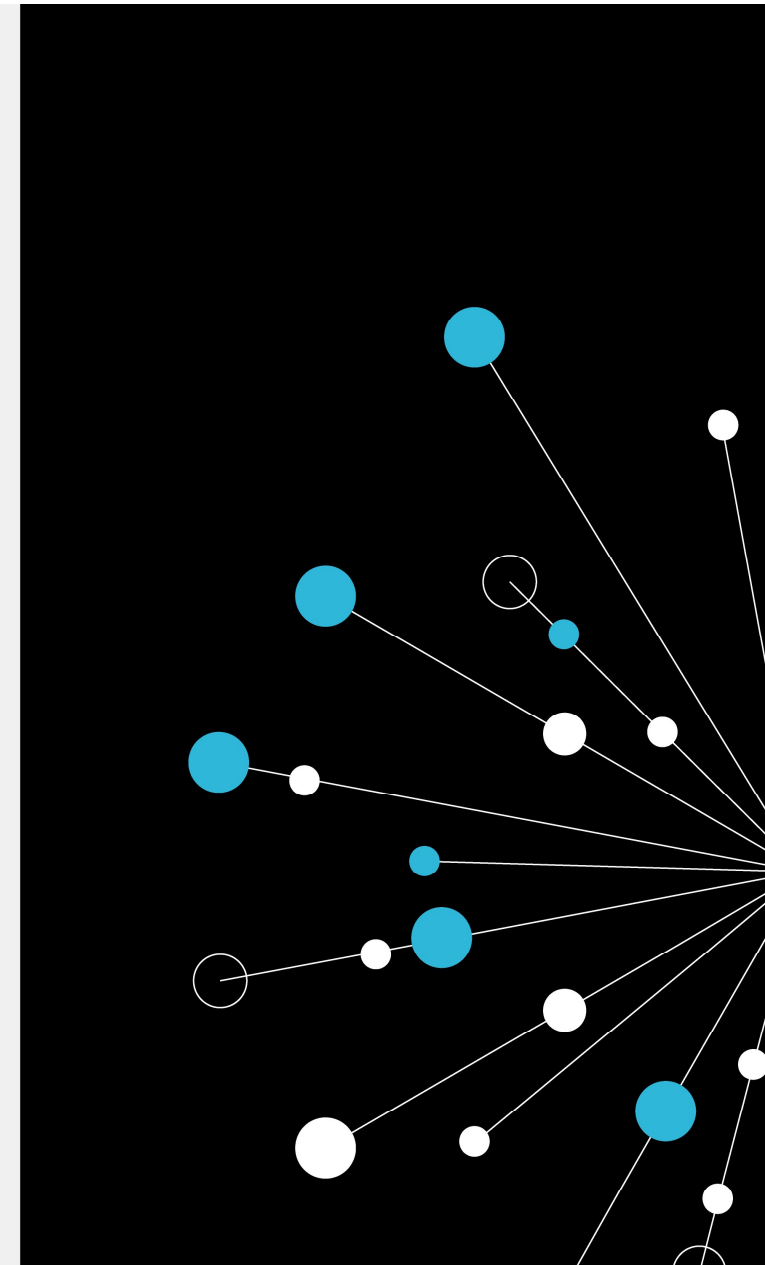
1. Important and urgent
2. Important but not urgent
3. Not important but urgent
4. Not important and not urgent

The Eisenhower matrix

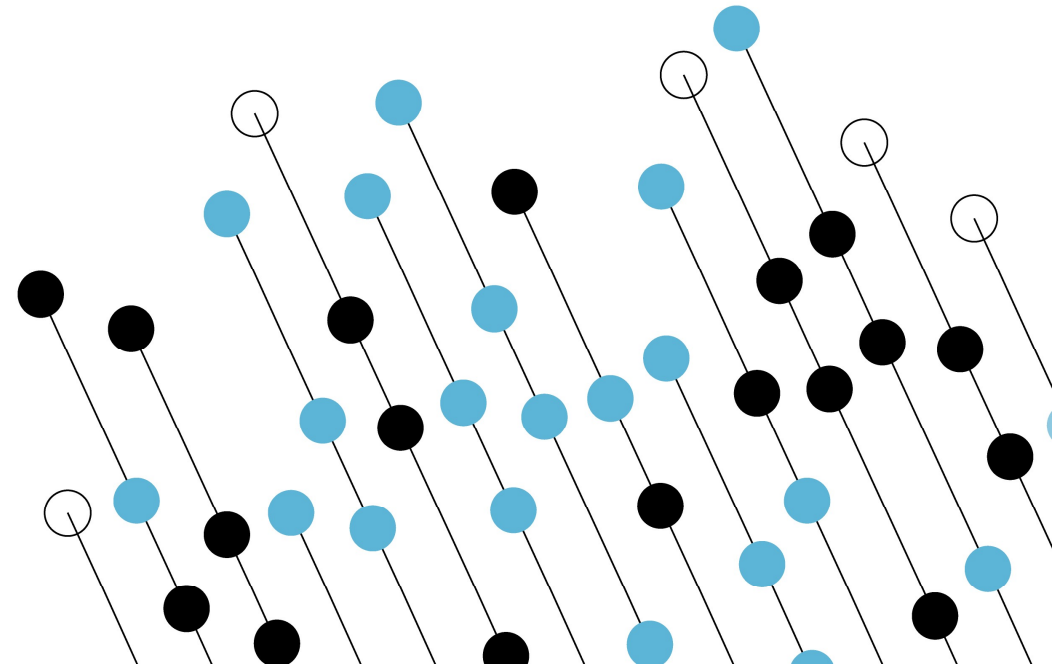
	Urgent	Not urgent
Important	Do now	Schedule for later
Not important	Delegate	Delete

Activity 7: Eisenhower Matrix

1. Write down all your assignments, projects, and tasks (work related and personal activities) that need to be done this week or month / year.
2. Create an Eisenhower matrix and place each task into the correct section based on its urgency and importance.



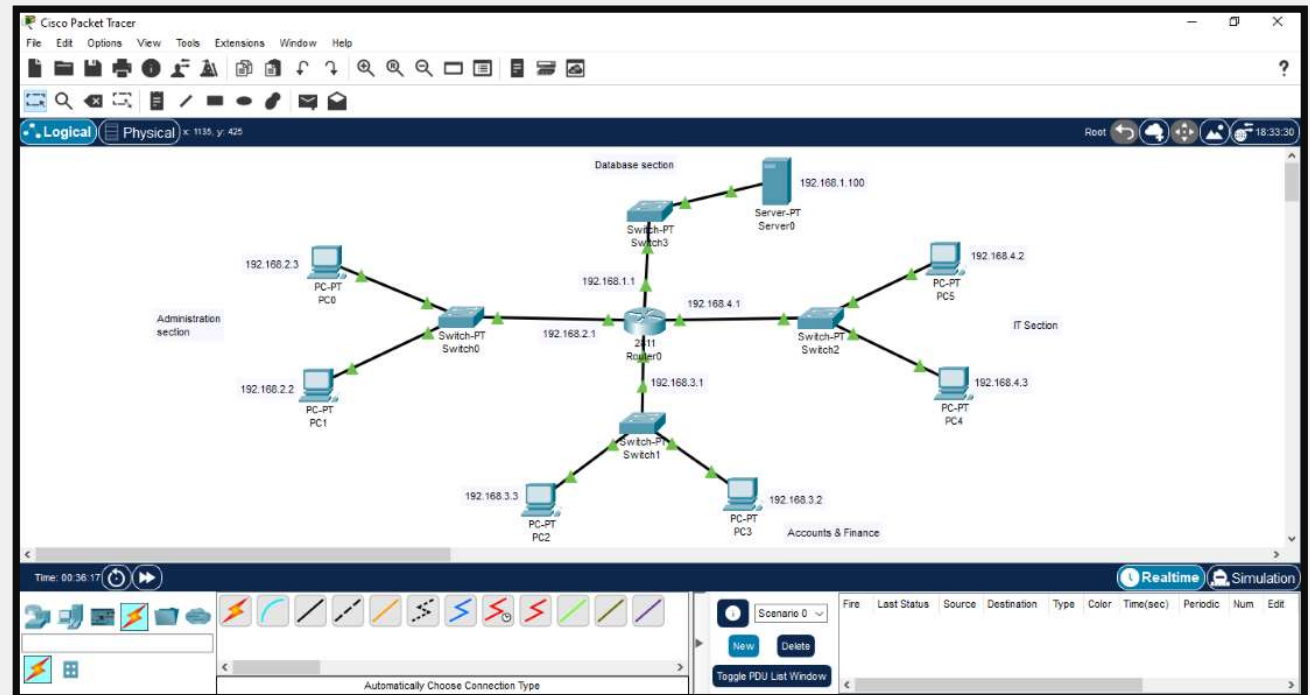
Introduction to Cisco Packet Tracer



Packet Tracer

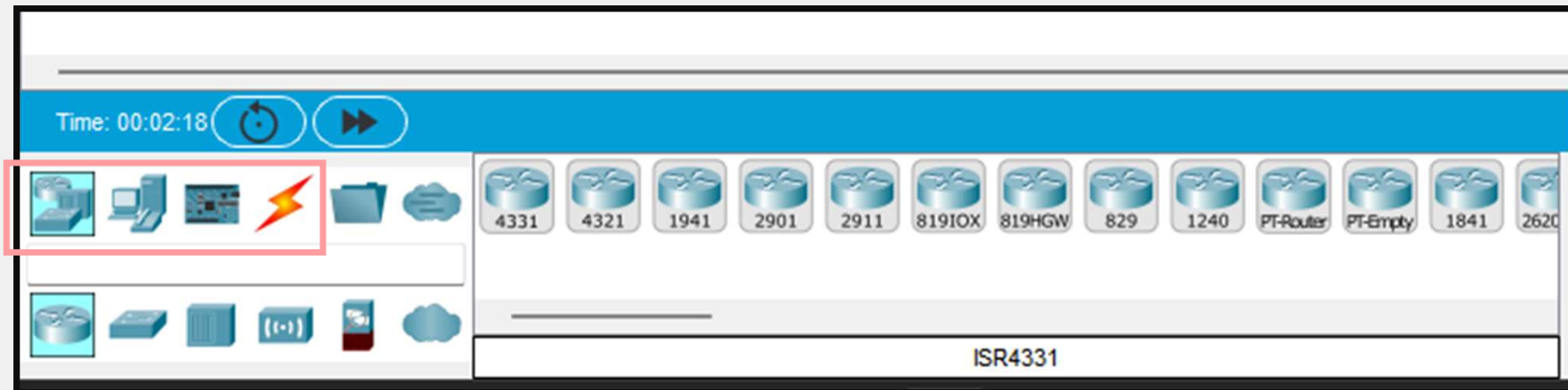
Packet Tracer is a network simulation tool developed by Cisco Systems.

It's an educational tool to help students and networking professionals learn about and practice network design, configuration, and troubleshooting.



Packet Tracer

- Devices can be found in the bottom left section, it will include items like desktops, servers, laptop, routers, switches, etc.
- You can then drag and drop any item on to the workspace above.

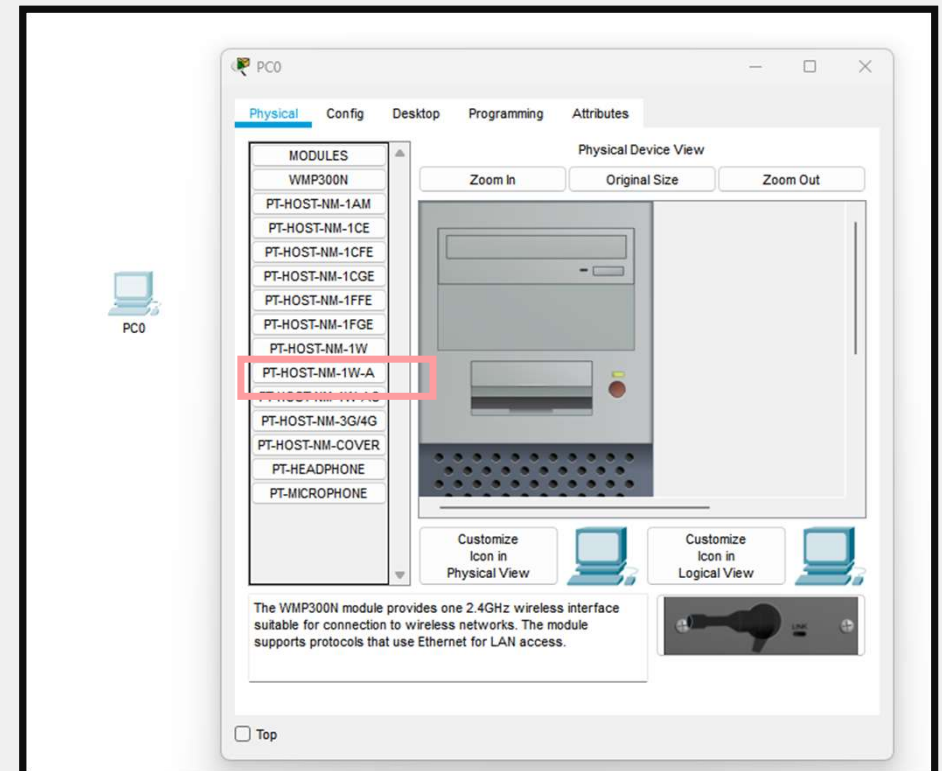


Packet Tracer

When you drag and drop an item in the workspace you can make further configurations when you click on it.

This includes:

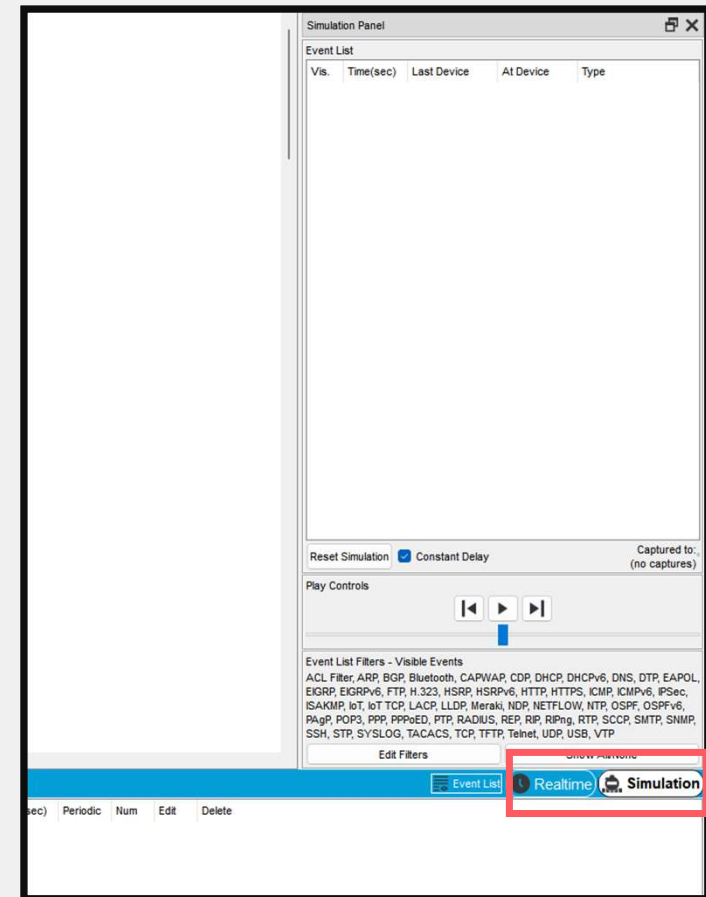
- installing / replacing physical hardware.
- changing the name or IP info of the device.
- simulating a desktop environment where you can run commands (e.g., connect via Wi-Fi, open a browser).



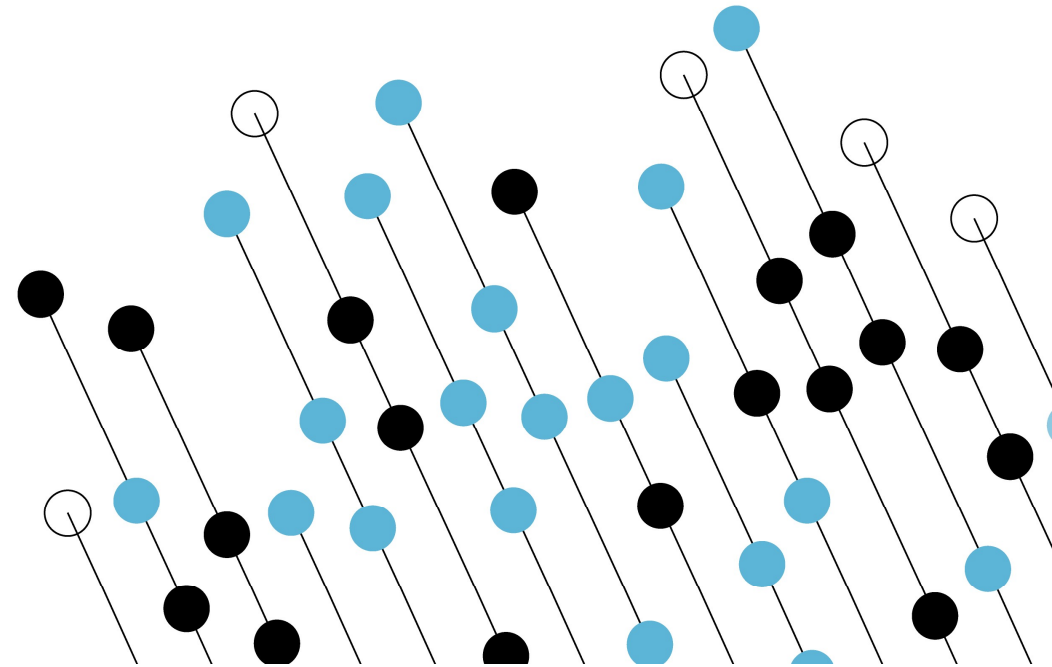
Packet Tracer

Simulation mode can be found in the bottom right section.

This is where you can inspect traffic flow and define what protocol filters you want to view.



Troubleshooting

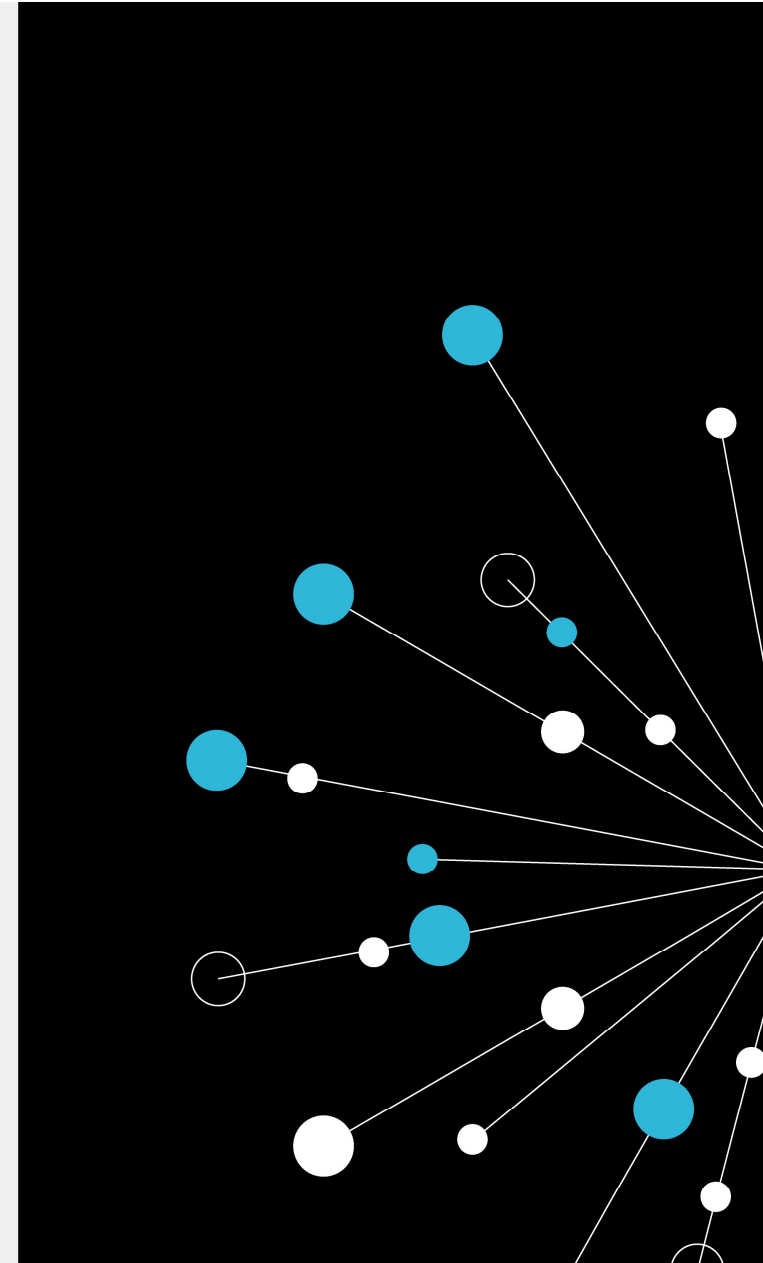


Troubleshooting activity

Now you can put your learning into practice.

Your tutor will set you a task where you are challenged to find a resolution to a problem and apply it in Packet Tracer.

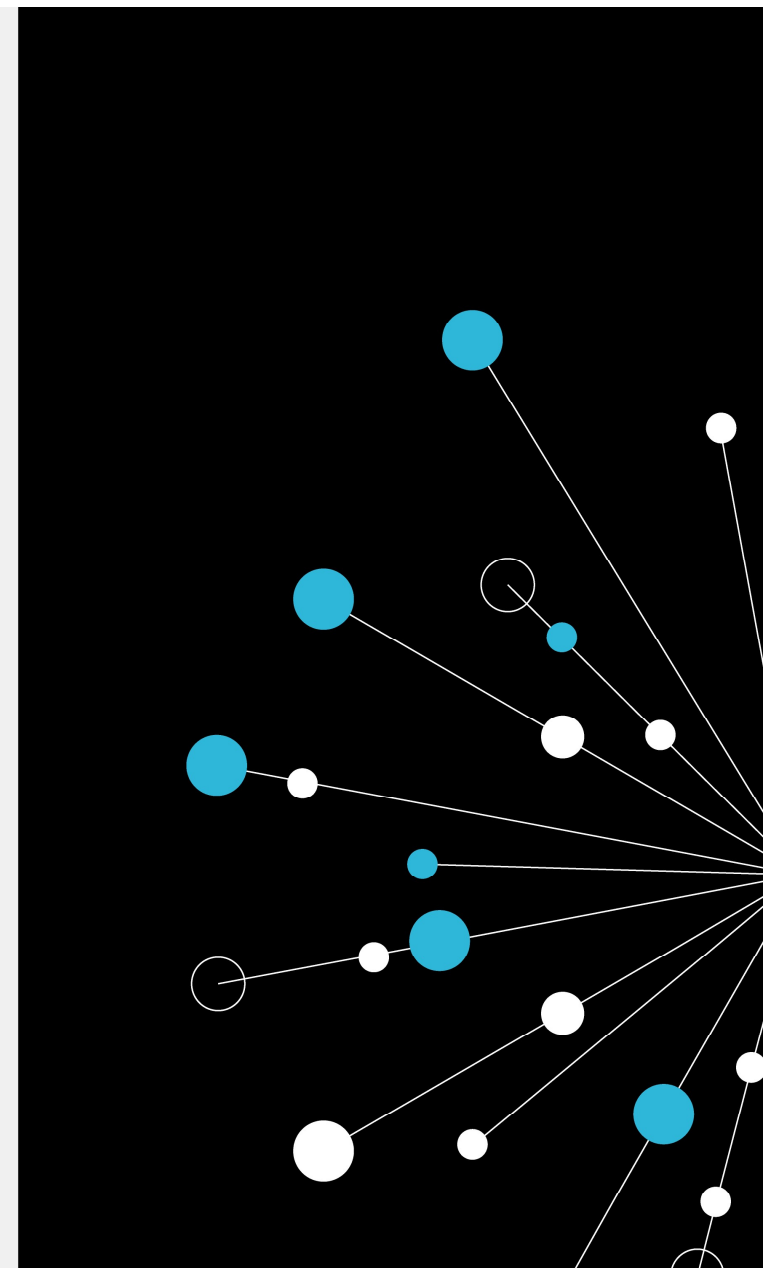
Once fixed, you need to document the process using the STARRS method.



Activity 8: Packet Tracer activity

Davis MacLean's laptop is unable to connect to the network.

1. Open Packet Tracer and the .PKA file for this activity. Apply your resolution to Davis MacLean's problem.
2. Using the STARRS method as a guide, write a short summary of how you tried to resolve the issue.



End of module review

- Diversity and cultural awareness
- Communication in the workplace
- Health and safety
- Numbering systems
- Technical documentation and problem solving
- Introduction to STARRS
- Task prioritisation
- Introduction to Cisco Packet Tracer
- Troubleshooting activity



End of module review

- Kindly Upload your completed Delegate Guide and *Certification of Completion to Bud under the following module:

Activity 1.3 | Discover and Practise | Core IT Principles: Legislation and IT Systems (Live Event)

- *Certification of Completion can be obtained by going to the live event module in Bud, selecting 'Start now', then ticking off the modules accordingly.

Learning instructions & materials

To complete this activity:

1. Attend a two-day live event (*), led by your tutor [| Start now |](#) - please don't start this learning path before the live event.
2. Submit your work in the 'Messages and Submissions' area above